

FACULTY OF SCIENCE & TECHNOLOGY

Second Semester B. Tech. (Civil/EC/ET/EN/EE/Mech./C.S.E./CE/IT/CT/CSE (DS)/CSE (AI)
CSE (AIML)/AI/AIML/AIDS/IOT/Robotics AI/Cyber Secu.) (C.B.C.S.) (NEP) Examination

CONSCIOUSNESS STUDIES**Time: [Three Hours]****[Maximum Marks: 70]****INSTRUCTIONS TO CANDIDATES**

1. All questions carry marks as indicated.
2. Solve Question 1 **OR** Question No. 2.
3. Solve Question 3 **OR** Question No. 4.
4. Solve Question 5 **OR** Question No. 6.
5. Solve Question 7 **OR** Question No. 8.
6. Solve Question 9 **OR** Question No. 10.

1. (a) Define Psychology. What is the subject matter of psychology? **7**
(b) Briefly elaborate the various subfields of psychology. **7**

OR

2. (a) Explain any two methods of psychology with its advantages and limitations. **7**
(b) Discuss any two schools of psychology in detail. **7**

3. (a) What are the cognitive processes? Elaborate the types of cognitive processes. **7**
(b) Explain the process of perception in detail. **7**

OR

4. (a) Illustrate any two principles of perceptual organization with neat sketches. **7**
(b) List the factors affecting the perceptual organization. **7**

5. (a) What is learning? Elaborate the conditions of observation learning. **7**
(b) Explain Pavlov's experiment on classical conditioning in detail. **7**

OR

6. (a) Explain B.F. Skinner's experiment of reinforcement and punishment. 7
(b) Explain the role of nature in classical conditioning. 7
-

7. (a) What is memory? Explain the three processes of memory in detail. 7
(b) Briefly describe the three key aspects of short-term memory. 7

OR

8. (a) What is sensory memory? Describe any two types of sensory memory in detail. 7
(b) Write a brief note on explicit memory. 7
-

9. (a) Explain the application of psychology in science and engineering. 7
(b) Write a brief note on any two senses. 7

OR

10. (a) Discuss the application of learning principles in advertising and education by giving suitable examples. 7
(b) What is forgetting? Briefly explain the interference theory of forgetting. 7

FACULTY OF SCIENCE & TECHNOLOGY

Second Semester B. Tech. (C.S.E. / IT / CT / CSE (DS) / CSE (CS) / CSE (AIML))
(C.B.C.S.) (NEP) Examination

MATHEMATICAL FOUNDATION OF COMPUTER SCIENCE

Time: [Three Hours]

[Maximum Marks: 70]

INSTRUCTIONS TO CANDIDATES

1. All questions carry marks as indicated.
2. Solve Question 1 **OR** Question No. 2.
3. Solve Question 3 **OR** Question No. 4.
4. Solve Question 5 **OR** Question No. 6.
5. Solve Question 7 **OR** Question No. 8.
6. Solve Question 9 **OR** Question No. 10.
7. Use of non-programmable calculator is permitted.

1. (a) Let $X = \{1, 2, 3, 4, 5, 6, 7\}$ and $R = \{(x,y) \mid x - y \text{ is divisible by } 3\}$. Show that R is an equivalence relation. 7
 (b) Let $A = \{1, 2, 3, 4\}$ and $R = \{(1,2), (2,3), (3,4)\}$ be a relation on A . Find transitive closure of R and draw its diagraph. 7

OR

2. (a) List all possible functions from the set $X = \{a, b, c\}$ to the set $Y = \{0, 1\}$. Indicate in each case whether the function is one-one, onto or both. 7
 (b) Prove that
 (i) $f_{A \cup B} = f_A + f_B - f_A \cdot f_B$
 (ii) $f_{A \cap B} = f_A \cdot f_B$
 Where f is characteristic function. 7

3. (a) If $U = \{a, b, c, d\}$. A and B are two fuzzy sets defined on U as given below:

	a	b	c	d
--	---	---	---	---

A	0.3	0.8	0.6	0.2
B	0.2	1.0	0.1	0.7

Find:

(i) $A \cup B$

(ii) $A = B$ and show that $(A \cup B)' = A' \cap B'$

7

(b) For the two sets A and B prove that:

(i) $A - (B \cup C) = (A - B) \cap (A - C)$

(ii) $A - (B \cap C) = (A - B) \cup (A - C)$

7

OR

4. (a) By the principle of mathematical induction, show that:

$$1^2 + 2^2 + 3^2 + \dots + n^2 = [n(n+1)(2n+1)] / 6, \text{ for } n \geq 1$$

7

(b) If R and S are fuzzy relations given by

$$R = \begin{matrix} & y_1 & y_2 \\ x_1 & [0.3 \ 0.1] \\ x_2 & [0.2 \ 0.8] \\ x_3 & [0.4 \ 0.6] \end{matrix} \quad S = \begin{matrix} & y_1 & y_2 \\ x_1 & [0.6 \ 0.2] \\ x_2 & [0.4 \ 0.8] \\ x_3 & [0.7 \ 0.9] \end{matrix}$$

Then find $R \cup S$, $R \cap S$, $R \circ S$.

7

5. (a) Fit a parabola $y = a + bx^2$ using the following data :

x	1	2	3	4	5
y	1.8	5.1	8.9	14.1	19.8

(b) Find the coefficient of correlation and equation of lines of regression using following data:

7

x	1	2	3	4	5
y	2	5	3	8	7

7

OR

6. (a) Fit a least square geometric curve $y = ax^b$ to the following data :

x	1	2	3	4	5
----------	---	---	---	---	---

y	0.5	2	4.5	8	12.5
---	-----	---	-----	---	------

(b) Obtain the rank correlation coefficient for the following data :

7

x	68	64	75	50	64	80	75	40	55	64
y	62	58	68	45	81	60	68	48	50	70

7

7. (a) Show that intersection of any two normal subgroups of a group G is a normal subgroup of G.

7

(b) Show that the set of matrices $A_\alpha = \begin{vmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{vmatrix}$, $\alpha \in \mathbb{R}$ form a monoid.

7

OR

8. (a) A non-empty subset of H of a group G is a subgroup of G iff:

(i) $a, b \in H \Rightarrow a * b \in H$

(ii) $a \in H \Rightarrow a^{-1} \in H$

7

(b) If f is a homomorphism of G into G', then:

(i) $f(e) = e'$ where e' is the identity element of G'

(ii) $f(x^{-1}) = [f(x)]^{-1}$, $\forall x \in G$.

7

9. (a) A box contains 5 red and 6 white marbles. In how many ways can 6 marbles be selected so that there are atleast two marbles of each colour?

7

(b) Find the closed form of generating function for the following :

(i) 3, -3, 3, -3, 3, -3,

(ii) 1, 0, -1, 0, 1, 0, -1, 0,

(iii) 0, 0, 1, 1, 1,

7

OR

10. (a) Solve the recurrence relation $a_{n+2} - 5a_{n+1} + 6a_n = 0$, $a_0 = 1$, $a_1 = -1$.

7

(b) Solve using Generating function $a_n = 3a_{n-1} + 2$, $a_0 = 1$.

7

FACULTY OF SCIENCE & TECHNOLOGY

Second Semester B. Tech. (C.S.E./IT/CT/CSE (DS)/CSE (CS)/CSE (AIML)/AIDS/IOT/Robotics AI/Cyber
Secu.) (NEP) Examination

ESSENTIALS OF PHYSICS

Time: [Three Hours]

[Maximum Marks: 70]

INSTRUCTIONS TO CANDIDATES

1. All questions carry marks as indicated.
2. Solve Question 1 **OR** Question No. 2.
3. Solve Question 3 **OR** Question No. 4.
4. Solve Question 5 **OR** Question No. 6.
5. Solve Question 7 **OR** Question No. 8.
6. Due credit will be given to neatness and adequate dimensions.
7. Assume suitable data wherever necessary.
8. Illustrate your answers wherever necessary with the help of neat sketches.
9. Use of non-programmable calculator is permitted.

List of Constants:

1. Plancks Constant $h = 6.626 \times 10^{-34}$ J.s
2. Mass of Electron $m_e = 9.11 \times 10^{-31}$ kg
3. Charge on Electron $e = 1.602 \times 10^{-19}$ c
4. Velocity of Light $c = 3 \times 10^8$ m/s
5. Mass of Alpha Particle $= 6.64 \times 10^{-27}$ kg
6. Mass of Proton $m_p = 1.67 \times 10^{-27}$ kg
7. Avogadro's Number $N_A = 6.023 \times 10^{26}$ atom/K mole

1. (a) Explain the Planck's hypothesis. Deduce the expressions of deBroglie wavelength for an electron accelerated through a potential difference of V volt. 5
 (b) Differentiate Bit and Qubit. Describe the term Quantum entanglement. 5
 (c) Calculate the de-Broglie wavelengths of 150 (gm) baseball moving with 35 (m/s) and an electron accelerated through 120 (V). 4

OR

2. (a) Show that the energy of a micro particle confined in an infinite one-dimensional potential well of length 'L' is quantized. 5
- (b) What is wave function (ψ)? Explain physical significance of wave function ψ . 5
- (c) Calculate the Uncertainty in position of a 50 (gm) bullet with velocity 30 (m/s) accurate to 0.01% and an electron with velocity 30 (m/s) accurate to 0.01%. 4
-
3. (a) Derive the mathematical expression for the numerical aperture and acceptance angle for optical fibre. 5
- (b) Classification of optical fibre on index profile. 5
- (c) An optical fibre has a NA of 0.2 and cladding of refractive index of 1.5. Determine the acceptance angle for the fibre in water which has refractive index of 1.33. 4

OR

4. (a) What is attenuation? Express the various mechanisms of attenuation in optical fibres. 5
- (b) Describe the application of optical fibre as a sensor. 5
- (c) Express attenuation in dB/Km, if 15% of the power fed at launching end of 1-km fibre is lost during propagation. 4
-

5. (a) Define Electrical Conductivity of metal and derive expression for Electrical Conductivity. 7
- (b) Discuss the classification of material on the basis of band gap. 5
- (c) Calculate the drift velocity of the free electrons with a mobility of $3.5 \times 10^{-3} \text{ m}^2/\text{V.s}$ in Copper for electric field strength of 0.5 V/m. 4

OR

6. (a) Draw the symbol of PNP and NPN Transistor. Explain the function of Emitter, Base and Collector. 2+3
- (b) What is Tunnel Diode with symbolic diagram? Draw V-I Characteristics of Tunnel Diode. 2+3

(c) Evaluate dc current gain for a PNP transistor in a common emitter mode if collector current is 2mA and base current is $20\text{ }\mu\text{A}$.

4

7. (a) Deduce the expression of Bethe's law. Discuss similarities and differences between Bethe's law and Snell's law.

6

(b) Show that electron describe a parabolic path when it enters in uniform transverse electric field.

4

(c) Determine the velocity of ions that pass undeflected through crossed E and B fields for which $E = 7.7\text{ kV/m}$ and $B = 0.14\text{ T}$.

4

FACULTY OF SCIENCE & TECHNOLOGY

Second Semester B. Tech. (CSE/ CE/ IT/ CT/ CSE (DS)/ CSE (CS)/ CSE (AIML))
(C.B.C.S.) (NEP) Examination

COMPUTER ARCHITECTURE AND ORGANIZATION**Time: [Three Hours]****[Maximum Marks: 70]****INSTRUCTIONS TO CANDIDATES**

1. All questions carry marks as indicated.
2. Solve Question 1 **OR** Question No. 2.
3. Solve Question 3 **OR** Question No. 4.
4. Solve Question 5 **OR** Question No. 6.
5. Solve Question 7 **OR** Question No. 8.
6. Due credit will be given to neatness and adequate dimensions.
7. Assume suitable data wherever necessary.
8. Illustrate your answers wherever necessary with the help of neat sketches.

1. (A) Explain the basic functional blocks of a computer system, including the Central Processing Unit (CPU), Memory, Input/Output Subsystems and Control Unit. **9**
(B) Define and explain different addressing modes used in a CPU. Discuss the significance of each mode. **5**

OR

2. (A) Discuss the concept of an instruction set in a CPU. **5**
(B) Explain the different components of Instruction Set architecture, including registers instruction execution cycle and addressing modes. **9**

3. (A) Describe the difference between fixed-point and floating point number representations. **5**
(B) Explain Booth's Multiplication algorithm. Discuss how it optimizes the multiplication of signed binary numbers and reduces the number of partial products compared to the traditional shift-and-add method. **9**

OR

4. (A) Explain how characters are represented in computers. Discuss common encoding schemes such as ASCII and unicode. 5
- (B) Compare and contrast the non-restoring and restoring division techniques used in binary division. 9
-

5. (A) Explain the concept of memory interleaving. Discuss its significance in improving memory access speed and how it is implemented in a system. 9
- (B) Discuss the relationship between cache size and block size in a cache memory system. 5

OR

6. (A) Discuss different types of cache mapping techniques, such as direct mapped, associative and set-associative mapping. 9
- (B) Explain the working of algorithms like Least Recently Used (LRU), First in First out and Optimal Page Replacement. 5
-

7. (A) Describe the different types of peripheral devices and their characteristics. Explain the distinction between input and output devices, providing examples. 5
- (B) Explain the concept of Software interrupts and exceptions. Illustrate with examples how software interrupts and exceptions are used in modern operating systems. 9

OR

8. (A) Explain how Direct Memory Access (DMA) works for data transfer between peripherals and memory without CPU intervention. 5
- (B) Explain how interrupts are used in process scheduling. Discuss the importance of interrupt handling in modern operating systems. 9

FACULTY OF SCIENCE & TECHNOLOGY

Second Semester B. Tech. (CS/ CE/ IT/ CD/ CSE (DS)/ CSE (CS)/ CSE (AIML))
(CBCS) (NEP) Examination

PYTHON PROGRAMMING**Time: [Three Hours]****[Maximum Marks: 70]****INSTRUCTIONS TO CANDIDATES**

1. All questions carry marks as indicated.
2. Solve Question 1 **OR** Question No. 2.
3. Solve Question 3 **OR** Question No. 4.
4. Solve Question 5 **OR** Question No. 6.
5. Solve Question 7 **OR** Question No. 8.
6. Solve Question 9 **OR** Question No. 10.

1. (A) Explain the working of input() function and eval() function in brief alongwith an example. **7**
(B) What are Datatypes in Python? Explain each datatype alongwith an example. **7**

OR

2. (A) Explain the importance of comments in Python Programming. Demonstrate with example how to use single-line and multi-line comments effectively. **7**
(B) Describe various types of tokens in Python. Explain each token type with an appropriate example. **7**

3. (A) Explain various arithmetic operators available in Python. Elaborate alongwith an example. **7**
(B) Distinguish between break statement and continue statement alongwith an example. **7**

OR

4. (A) Write a program in Python to check whether the given number is odd or even. **7**

(B) Describe various decision-making statements in Python. Design a Python program using nested if-else block. 7

5. (A) Explain syntax and basic structure of a function in Python. Write a program in Python to perform addition of two numbers using user-defined function. 7
- (B) Difference between local and global variables alongwith an example. 7

OR

6. (A) Describe the role of return statement in a Python function. Illustrate the working of return statement alongwith an example. 7
- (B) Describe the syntax and usage of lambda functions in Python. Compare lambda functions with normal functions using examples. 7
-

7. (A) Explain how tuples are created in Python. Illustrate different ways of creating a tuple with an example. 7
- (B) Write a program in Python that performs different operation on a tuple. 7

OR

8. (A) Explain the working of zip() functions in Python alongwith an example. 7
- (B) Describe 'in' and 'not in' operators in sets. Illustrate with an example. 7
-

9. (A) Construct a program in Python to write a text into a file. 7
- (B) Explain how numeric data can be written to a file in Python. Demonstrate with a program. 7

OR

10. (A) Design a program in Python to read integers line by line and calculate their sum. 7
- (B) Write a program in Python where new text is added without overwriting existing content. 7